Machine Learning Prediction of Exoplanet Equilibrium Temperature

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1 Introduction

The objective of this project is to use machine learning techniques to predict the equilibrium temperature of exoplanets based on their observed physical and orbital properties. The equilibrium temperature of a planet is a fundamental astrophysical quantity, as it provides a first-order estimate of the planet's thermal environment and potential habitability. It is primarily governed by the amount of stellar radiation received, as well as orbital characteristics and stellar properties. The equilibrium temperature can be approximated by

$$T_{eq} = \left(\frac{(1-A)L_*}{16\pi\sigma a^2} \right)^{1/4} \tag{1}$$

where A is the Bond albedo, L_* is the stellar luminosity, a is the semi-major axis, and σ is the Stefan–Boltzmann constant. This relationship highlights the strong dependence of temperature on incident stellar radiation.

The dataset used in this analysis consists of a collection of confirmed exoplanets from the NASA Exoplanet Archive with measured planetary and stellar parameters. The target variable for this study is the planetary equilibrium temperature (pl_eqt), while the input features include orbital properties such as semi-major axis and eccentricity, planetary characteristics like mass and radius, and stellar properties including mass, temperature, and density.

To better understand the dataset, several exploratory visualizations were generated. The distribution of equilibrium temperatures, shown in Figure 1, reveals a broad range spanning from a few hundred to over 2000 K, with a noticeable concentration of planets between roughly 500 K and 1700 K. This spread reflects the diversity of exoplanet environments, from relatively temperate worlds to highly irradiated hot planets.

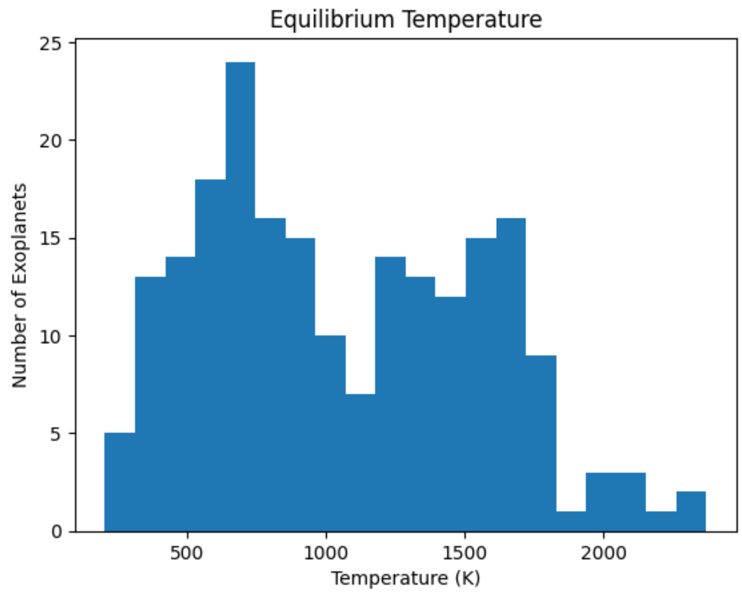

Figure 1: Histogram of exoplanet equilibrium temperatures. The distribution spans from a few hundred Kelvin to over 2000 K, with a concentration between approximately 500 K and 1700 K, reflecting the diversity of planetary environments.

Further insight is gained by examining how equilibrium temperature relates to key features. Figure 2 shows equilibrium temperature as a function of insolation flux. A strong, nonlinear relationship is clearly visible: as insolation increases, equilibrium temperature rises in a smooth, curved trend. This is consistent with physical expectations, since temperature scales approximately with the fourth root of incident flux F.

$$T_{eq} \propto F^{1/4} \tag{2}$$

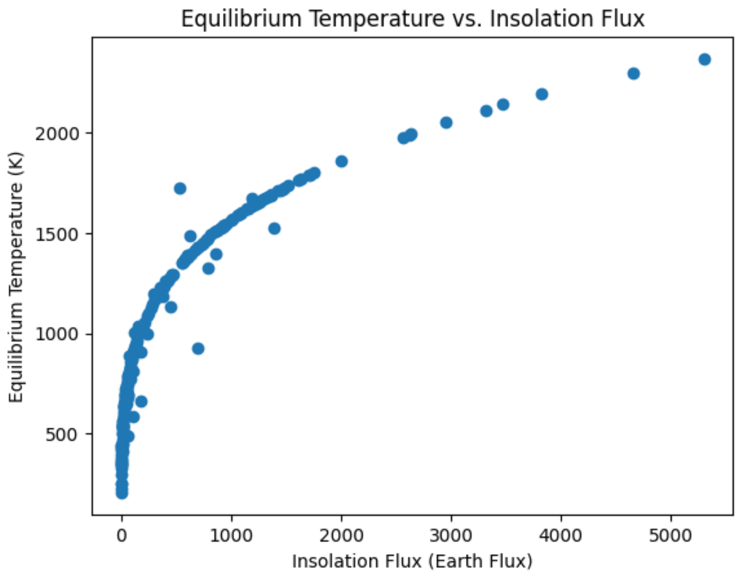

Figure 2: Equilibrium temperature as a function of stellar insolation flux. A strong nonlinear correlation is observed, consistent with theoretical expectations that temperature scales with the fourth root of incident flux.

In contrast, Figure 3 shows equilibrium temperature versus orbital eccentricity. Here, no strong correlation is evident; most planets cluster at low eccentricity values, and temperature varies widely within that region. This suggests that eccentricity plays a comparatively minor role in determining equilibrium temperature within this dataset.

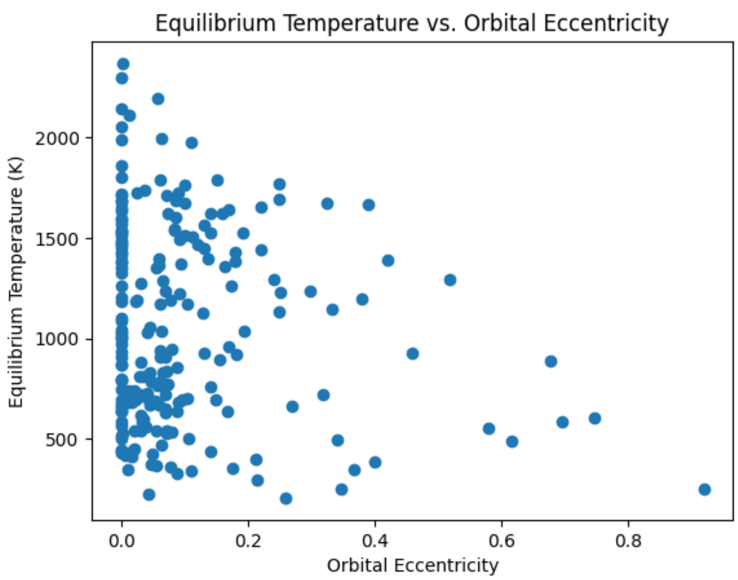

Figure 3: Equilibrium temperature versus orbital eccentricity. Most planets cluster at low eccentricities, and no strong correlation is evident, suggesting eccentricity plays a minor role.

Additional relationships were explored with stellar properties. As shown in Figure 4, there is a moderate positive correlation between stellar mass and equilibrium temperature, reflecting the fact that more massive stars tend to be more luminous. Meanwhile, Figure 5 shows a weaker and more scattered relationship with stellar density, indicating that density is not a primary driver of planetary temperature.

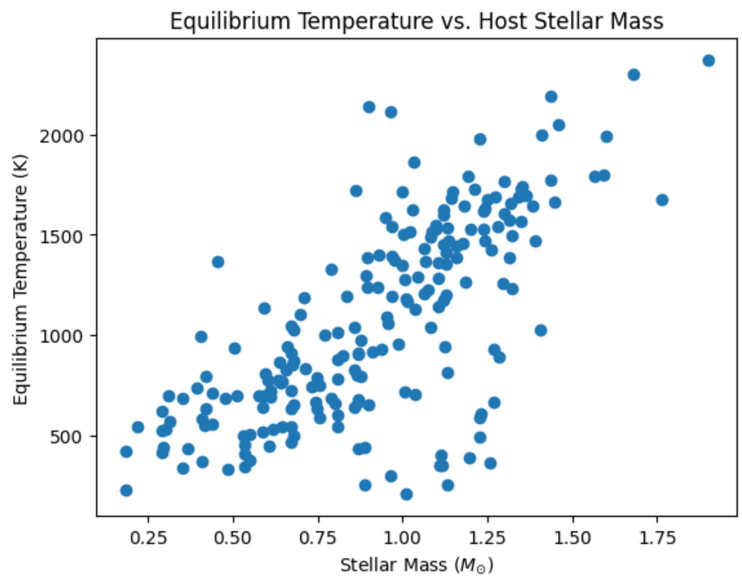

Figure 4: Equilibrium temperature versus stellar mass. A moderate positive correlation is observed, indicating that planets around more massive stars tend to be hotter.

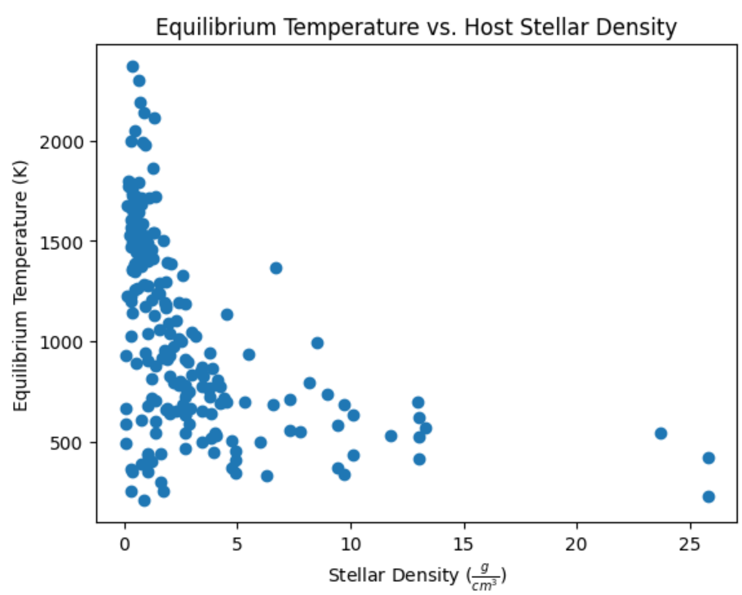

Figure 5: Equilibrium temperature versus stellar density. The relationship is weak and scattered, indicating that stellar density is not a strong predictor.

Overall, this exploratory analysis suggests that while several features contribute to equilibrium temperature, insolation flux is likely to be the dominant predictor. This observation motivates the use of machine learning models capable of capturing nonlinear relationships between features and the target variable.

2 Approach

To model the relationship between planetary properties and equilibrium temperature, two ensemble learning algorithms were implemented: a Random Forest Regressor and an XGBoost Regressor. Both methods are well-suited for tabular astrophysical data, as they can capture nonlinear dependencies and interactions between features without requiring extensive preprocessing.

The input features for both models consist of a subset of planetary and stellar parameters, including orbital period, semi-major axis, planetary mass and radius, density, eccentricity, insolation flux, inclination, and several stellar properties such as effective temperature and mass. The target variable is the equilibrium temperature, making this a supervised regression problem.

The Random Forest model was configured with 200 trees and a limited number of features considered at each split. This approach reduces variance by averaging predictions across many decision trees, making it robust to noise and overfitting. In contrast, the XGBoost model uses gradient boosting, where trees are added sequentially to correct errors made by previous trees. This allows it to achieve high predictive accuracy, particularly for structured datasets like the one used here.

Model performance was evaluated using several metrics, including the coefficient of determination (R^2), root mean squared error (RMSE), and mean absolute error (MAE). The equations for these are as follows:

$$R^2 = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2} \tag{3}$$

$$RMSE = \sqrt{\frac{1}{n}\sum(y_i - \hat{y}_i)^2} \tag{4}$$

$$MAE = \frac{1}{n}\sum|y_i - \hat{y}_i| \tag{5}$$

These metrics provide complementary information: $\mathbf{R^2}$ measures how well the model explains variance in the data, while RMSE and MAE quantify prediction error in physically meaningful units (Kelvin). To ensure reliable performance estimates, 5-fold cross-validation was employed.

3 Initial Results

The initial models, using baseline hyperparameters, already demonstrated strong predictive performance. This is evident in the predicted versus true value plots.

Figure 6 shows the Random Forest predictions. The data points lie close to the diagonal line, indicating that the model is able to capture the overall trend in the data. However, there is some scatter, particularly at higher temperatures, suggesting room for improvement.

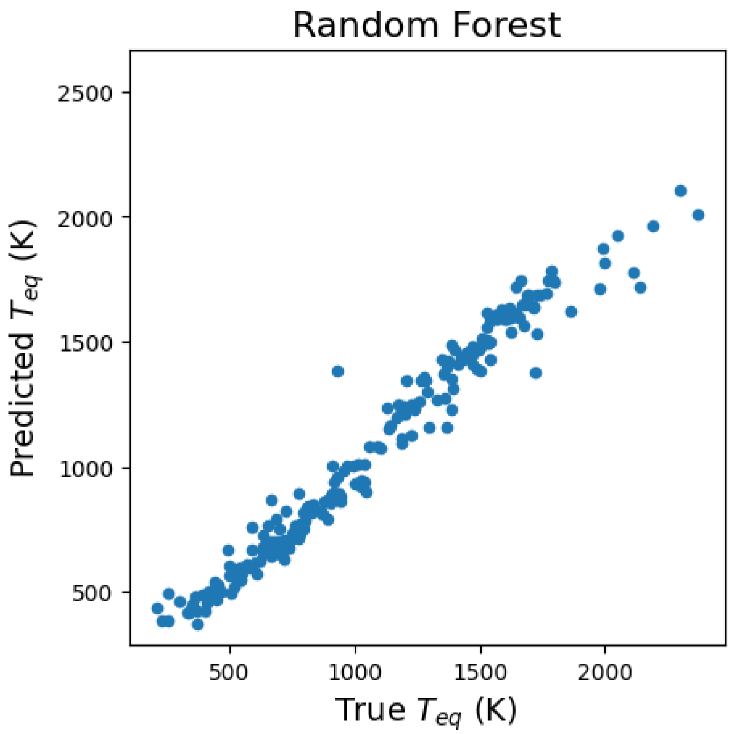

Figure 6: Random Forest predicted versus true equilibrium temperatures. Points lie close to the diagonal, indicating good performance, though some scatter remains at higher temperatures.

Figure 7 shows the XGBoost predictions, which appear even more tightly clustered around the diagonal. This indicates that XGBoost is achieving slightly better predictive accuracy than the Random Forest model, likely due to its boosting framework.

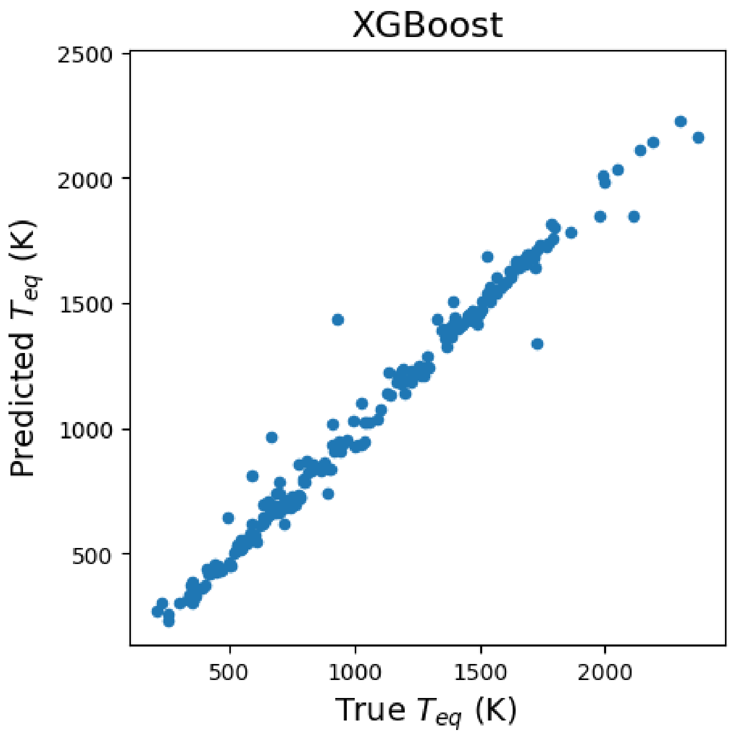

Figure 7: XGBoost predicted versus true equilibrium temperatures. The tighter clustering around the diagonal indicates improved predictive accuracy compared to Random Forest.

Feature importance analysis provides further insight into model behavior. As shown in Figure 8, insolation flux overwhelmingly dominates all other features in terms of importance. This aligns with the earlier exploratory analysis and confirms that the models are learning physically meaningful relationships.

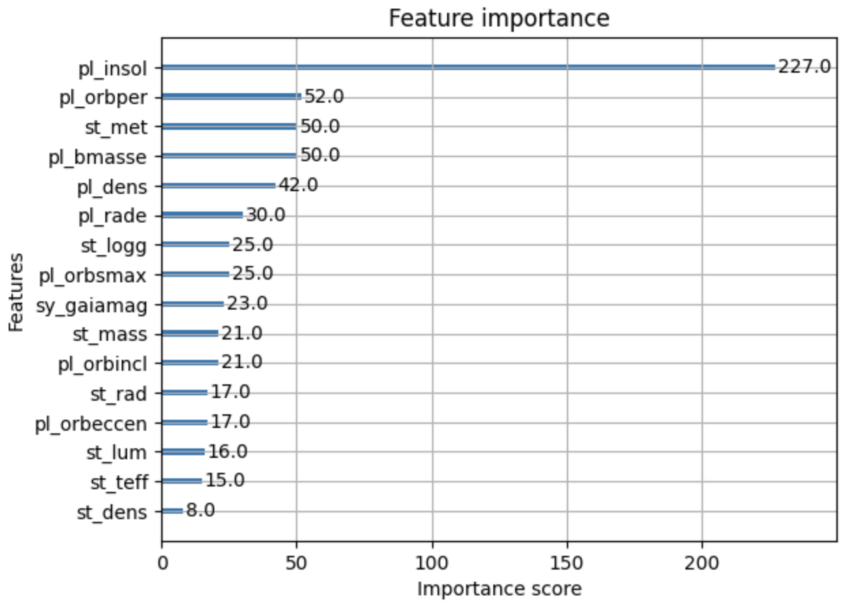

Figure 8: Feature importance from the XGBoost model. Insolation flux dominates all other features, confirming it as the primary driver of equilibrium temperature.

These initial results indicate that the training dataset is sufficient for capturing the primary relationships governing equilibrium temperature. However, the dominance of a single feature suggests that the model may be overly reliant on insolation flux, motivating further investigation.

3.1 Initial Model Performance

The baseline performance of the Random Forest and XGBoost models is summarized in Table 1.

Model	Train R^2	Test R^2	RMSE (K)	MAE (K)
Random Forest	0.994 ± 0.001	0.959 ± 0.014	82.35	52.41
XGBoost	1.000 ± 0.000	0.977 ± 0.017	38.01	30.24

Table 1: Performance of baseline models.

Both models achieve high predictive performance, with test R^2 values exceeding 0.95, indicating that the models explain the vast majority of variance in equilibrium temperature. The XGBoost model outperforms the Random Forest model, achieving both higher R^2 and significantly lower RMSE and MAE values.

The near-perfect training R^2 values, particularly for XGBoost, suggest that the models are highly expressive and capable of closely fitting the training data. However, the small gap between training and test performance indicates that overfitting is limited and that both models generalize well.

The RMSE values indicate that the typical prediction error is on the order of tens of Kelvin, with XGBoost achieving particularly low error. The relatively small difference between RMSE and MAE suggests that large outliers are not dominant, and that prediction errors are fairly consistent across the dataset.

4 Final Results and Reflection

4.1 Effect of Hyperparameter Tuning

To improve model performance and better understand feature contributions, several refinements were explored. Hyperparameter tuning was performed using grid search, which systematically tested combinations of model parameters to identify optimal settings. This process resulted in modest but consistent improvements in performance metrics, as shown in Table 2.

Model	RMSE (K)	MAE (K)
Random Forest	29.64	23.43
XGBoost	38.56	31.01

Table 2: Model performance after hyperparameter tuning.

Hyperparameter tuning leads to a substantial improvement in the performance of the Random Forest model, reducing both RMSE and MAE by more than a factor of two. In contrast, the XGBoost model shows little improvement and slightly worse performance compared to its baseline configuration.

This suggests that the initial XGBoost model was already near optimal for this dataset, while the Random Forest model benefited significantly from tuning. After optimization, the Random Forest model achieves comparable or better performance than XGBoost in terms of error metrics.

4.2 Effect of Feature Selection

Feature selection was also applied to reduce redundancy and improve interpretability. Highly correlated features were removed, and low-importance features were excluded based on model-derived importance scores. These steps simplified the model without significantly degrading performance. The performance after initial feature selection is shown in Table 3

Model	RMSE (K)	MAE (K)
Random Forest	40.53	31.60
XGBoost	45.39	36.03

Table 3: Model performance after feature selection.

Feature selection results in a degradation of model performance for both models, with increases in both RMSE and MAE. This indicates that some of the removed features, while potentially redundant, still contained useful information for predicting equilibrium temperature.

The performance drop suggests that the models benefit from having access to a broader set of features, even if some are correlated, highlighting the ability of tree-based methods to handle feature redundancy effectively.

4.3 Effect of Removing Insolation Flux

A particularly important experiment involved removing the insolation flux feature. When this feature was excluded, model performance dropped substantially (shown in Table 4), confirming that it plays a central role in determining equilibrium temperature. While additional feature engineering partially recovered performance (shown in Table 5, the results remained inferior to models that included insolation flux.

Model	RMSE (K)	MAE (K)
Random Forest	111.76	87.60
XGBoost	84.35	66.85

Table 4: Model performance after removing insolation flux.

Model	RMSE (K)	MAE (K)
Random Forest	122.00	83.25
XGBoost	75.12	58.63

Table 5: Performance after feature engineering without insolation flux.

Removing the insolation flux feature leads to a dramatic decrease in model performance, with RMSE and MAE increasing by more than a factor of two. This confirms that insolation flux is the dominant predictor of equilibrium temperature, consistent with theoretical expectations.

The significant performance degradation highlights the extent to which the models rely on this physically meaningful feature. Without it, the models must infer temperature indirectly from other variables, resulting in substantially reduced accuracy.

This highlights an important trade-off: while machine learning models can identify dominant predictors, those predictors often reflect underlying physical laws. In this case, the strong dependence on insolation flux is not a limitation of the model, but rather a reflection of the physics governing planetary temperatures.

4.4 Final Cross-Validation Performance

Model	Train R^2	Test R^2
Random Forest	0.989 ± 0.001	0.924 ± 0.016
XGBoost	1.000 ± 0.000	0.957 ± 0.004

Table 6: Cross-validation performance of final models.

The final cross-validation results indicate that both models maintain strong predictive performance, though with slightly reduced R^2 values compared to the initial models. The Random Forest model shows a larger drop in performance, suggesting increased sensitivity to feature modifications.

The XGBoost model remains more robust, retaining a high test R^2 with low variance across folds. This indicates that it generalizes well and is less affected by changes in the feature set.

Overall, both Random Forest and XGBoost proved to be effective for this task, with XGBoost showing slightly superior performance. The models successfully captured the nonlinear relationships present in the data and produced accurate predictions across a wide range of temperatures.

5 GitHub Repository

The full project is available at: `https://github.com/clairemol/final-project`

The repository includes the Jupyter notebook, dataset, and documentation describing the workflow and results.

6 AI Statement

AI tools were only used to assist with the removing of insolation flux and the evaluation of that model. AI helped come up with the engineered features used for the model, but the relationships and math done were done by hand.

All packages used were either used in class or related to a topic covered in class (like xgboost), so there was no new information to be learned from scikit-learn documentation.